Jonathan Hampton

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Experience

AI Engineer, Cahaba Federal Solutions — Huntsville, AL
Jan. ’24 - Present

- Lead AI engineer on the U.S. Army’s $15M Autonomous Aimpoint Selection and Tracking (A2ST) program, developing a system-agnostic edge AI platform for real-time autonomous perception and control.
- Architect end-to-end ML pipelines spanning Blender-based synthetic data generation, multi-node distributed PyTorch training, and model optimization for server and edge accelerators.
- Profile training and inference workloads using PyTorch Profiler and NVIDIA Nsight to identify compute/memory bottlenecks, optimizing performance across training and deployment hardware.
- Prototype novel AI capabilities for electro-optical and infrared (EO/IR) imagery, including real-time 3D shape estimation, 6-DoF differentiable rendering–based pose estimation, and sim-to-real fidelity enhancement using vision-language-action world models.
- Served as primary developer of a C++ ONNX-to-Versal AI Engine compiler for AMD/Xilinx hardware, implementing distributed Conv2D and ReLU kernels across FP32, FP16, and INT8.
- Collaborate with hardware, optical, and systems engineers to integrate the A2ST platform into several fielded systems.

Machine Learning Engineer, Radiance Technologies — Huntsville, AL
May ’21 - Jan. ’24

- Developed real-time neural object detection and tracking models for the U.S. Army’s Affordable Laser Weapon Module (ALWM), outperforming traditional tracking algorithms across two successful field demonstrations.
- Built and optimized a multithreaded C++ neural network inference runtime; combined architecture design, structured pruning, quantization, and runtime optimization to achieve 1,000+ FPS for FPGA deployment.
- Designed a modular OpenGL simulation platform used across three funded programs to generate synthetic imagery for neural network training and algorithm evaluation.
- Built and maintained a five-node on-premises compute cluster for distributed neural network training using Docker, Kubernetes, and Ray.

Technical Skills

Languages: Python, C++, CUDA

Training & Deployment: PyTorch, Lightning, Ray, ONNX, HuggingFace, Hydra, TensorRT, OpenVINO, Vitis

Optimization: PyTorch-Profiler, ModelOpt, Nsight, Triton

Data & Visualization: H5py, Pandas, CVAT, Plotly/Dash, Matplotlib

Vision & Graphics: OpenCV, TorchVision, PyTorch3D, Blender, NVDiffRast, OpenGL, PIL, Scikit-Image

Reinforcement Learning: Gymnasium, TorchRL, RLlib, Isaac-Lab

Systems: Git, Docker, Kubernetes, CMake, Jenkins, Linux, MLFlow

Misc: Azure/AWS/GCP, Protobuf, QEMU, Vivado, Yocto, VSCode

Education

MS **Computer Science**, Georgia Institute of Technology — 2021 - 2025
- Specialization: Machine Learning, 3.8/4.0 GPA

BS **Mathematics**, East Tennessee State University — 2014 - 2020
- Specialization: Computer Applied Mathematics

Selected Projects

LitDreamerV3 (2023, 2026)

- Reimplemented Google DeepMind’s DreamerV3 world-model reinforcement-learning algorithm from PyTorch primitives, including the learned latent dynamics model, actor-critic learning, replay, and latent imagination across multiple control domains.

LitWorldModels (2022)

- Reimplemented World Models in PyTorch Lightning, combining a variational autoencoder for visual compression, an MDN-RNN for probabilistic latent dynamics, and a CMA-ES controller for CarRacing. Structured data collection, model training, and controller optimization as separate stages.

Selected Publications

Exploring Interplay between Simulations and Generative AI through Case Studies — Oct. 2023
AlaSim Modeling and Simulation Conference
Jonathan Hampton, Blake Anderton Ph.D.

- Implemented and evaluated generative-modeling approaches including a learned world model for sensor-control agent training and ReCycleGAN for temporally consistent simulation-to-real style transfer.

Tracking Enhancements from AI Methodologies Demonstrated on ALWM — Apr. 2024
National Fire Control Symposium
Blake Anderton Ph.D., Jonathan Hampton